

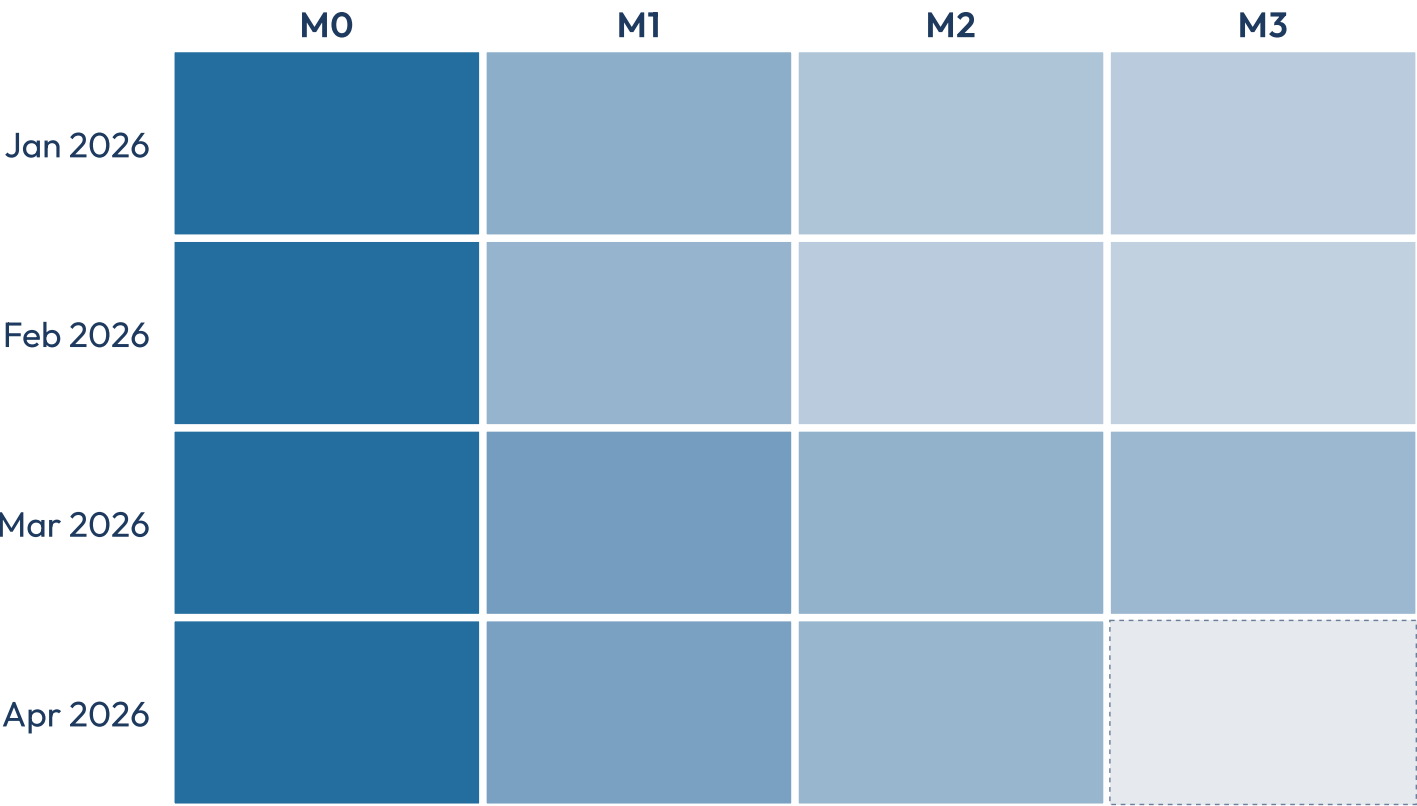
HEATMAP · A NUMERIC MATRIX READ AS INTENSITY

Where does it concentrate?

The chart family could compare magnitudes, decompose a total, plot a trend and score a 2×2 — but it had no way to put a **number at every crossing of two dimensions**. Cohort retention, a risk matrix, usage by hour and weekday: the shape a deck reaches for when the finding is a pattern rather than any single value.

RETENTION · 2026 COHORTS

Retention decays fastest in month two.



A missing cell is not a zero.

April has no month-three number because April has not reached month three.

That is the **normal** state of a cohort table, not an authoring error — so the crossing paints the bare neutral with a dotted edge and carries no intensity at all.

Pad it with  instead and the chart states a measurement nobody took, at the bottom of the ramp, where it reads as a collapse.

The ramp is the choropleth's, not a new one.

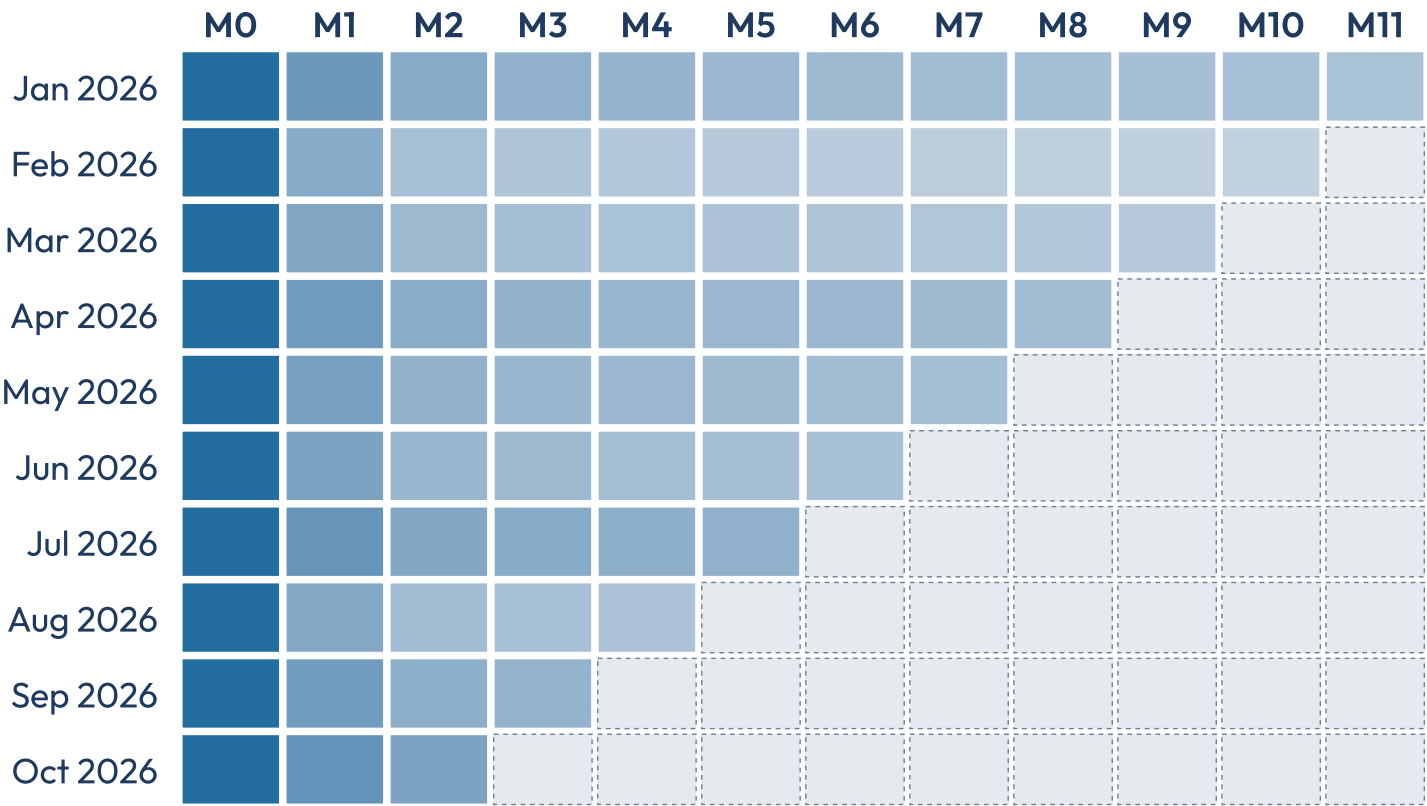
A heatmap and a `map` encode the same thing — one continuous value as one intensity — differing only in whether the cells are a grid or a geography. So this takes the mechanism `map` already proved: one hue mixed into a **neutral anchor**, never into the page. Mixing toward the canvas looks identical on light and inverts on dark, where a low value sinks *below* the empty cell it is supposed to sit above.

The cell carries no number, and that was measured.

Text on a colored fill is the one placement this family will not guarantee contrast for. The heatmap tried to buy an exception — print the value, flip its ink partway along the ramp — and the exception did not survive the palette sweep: across all **33 palettes** on both canvases, the best single crossover still leaves **318 cells** below AA, worst `2.60:1`. A per-palette crossover clears it but cannot be *expressed*, because a slide flips its own canvas and CSS has no numeric `light-dark()`. The intensity carries the reading; the numbers stay in the description a screen reader hears.

RETENTION · THE CEILING

Ten cohorts by twelve months.



Ten by twelve – the ceiling, with a ragged tail

When not to reach for it.

Intensity is read approximately — that is its strength for a pattern and its weakness for a comparison. One row of numbers is not a matrix; it is a `bar`, where length is judged precisely. Cells that are verbs or owners belong in `matrix-grid`. A value per country belongs on `map`. And if the decision turns on which of two similar cells is larger, the value belongs on an axis.